(Please write your Enrollment Number) Enrollment No. 06011720221

MID-TERM EXAMINATION
(Course Name : B. Tech. CSE/IT/CSE-AI) (Semester : 6th)
(March, 2024) OFF LINE mode

Subject Code: BCS 306	**Subject: Compiler Design**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Attempt all Questions	

Q1	Attempt all parts	**(2.5*4)**	
	(a) Define CFG? Write about Derivation trees and Parse tree with suitable example.	UNIT-1	
	(b) Discuss about role of Lexical Analyzer? What is input buffering?	UNIT-1	
	(c) Define FIRST and FOLLOW functions and Compute FIRST and FOLLOW for the following grammar: S→(L) /a L→ LS/S	UNIT-2	
	(d) Discuss Recursive Descent Parser? Apply the parser on following grammar. A→BC B→C C→c	UNIT-2	
Q2	**(Attempt any Two Parts) UNIT-1**		**(5,5)**
	(a) Explain various phases of a compiler with neat diagram?		
	(b) Discuss about bootstrapping and it's use with a neat diagram?		
	(c) Define Ambiguity? Check whether the following grammar *Ambiguous or not?* (i) S→SS/AB A→Aa/a B→Bb/b (ii) S→AA A→aA A→b		
Q3	**(Attempt any Two Parts) UNIT-2**		**(5,5)**
	(a) Discuss the difference between Top-down Parsing and Bottom up Parsing? Write about Operator precedence parser with an example.		
	(b) Construct SLR Parsing table for the grammar S→L=R/R, L→*R/id, R→L		
	(c) What is shift reduce parser, and parse the input string **int id,id;** using shift-reduce parser using the following grammar: S→TL; T→int | float L→L,id | id		